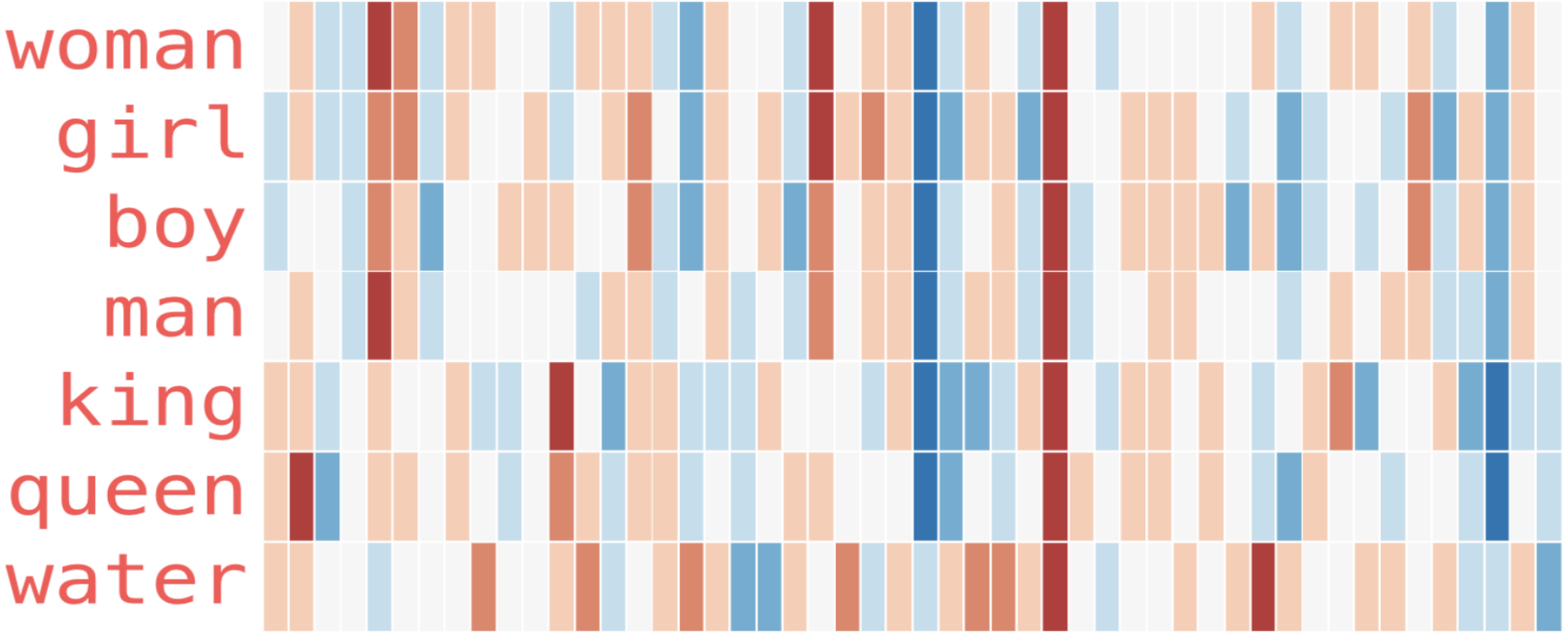
AI Technical Concepts

How an LLM (e.g.GPT) Works

- “Thinks” in patterns of 12,800 numbers each.
- Combines words into phrases, phrases into sentences, sentences into concepts.
- Predicts next “thought” and chooses best next word.
- 12 “matrix” equations, one page of code, describes the essential mechanism

Word/Idea “Embedding”



“Patterns”

- AI identifies concepts by number sequences
  - Word embedding
  - Photo by pixel “grayscale” values
  - Sound by frequencies at each time point
  - Color by red/green/blue values

High-Dimensional Spaces

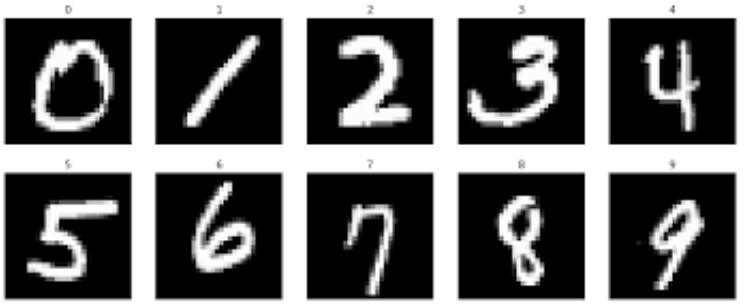
- Patterns are points in high-dimensional space
  - Two-number pattern is a position in 2-D space, e.g. latitude/longitude, or x/y
  - Three-number pattern is a point in 3-D space, e.g. latitude/longitude/altitude
- “Hypercube” sequence
  - Line, square, cube, …
  - Number of corners as dimension d rises?
- High-dimensional space is **big**
  - Largest distance corner-to-corner
  - Proportion of inscribed “hypersphere”

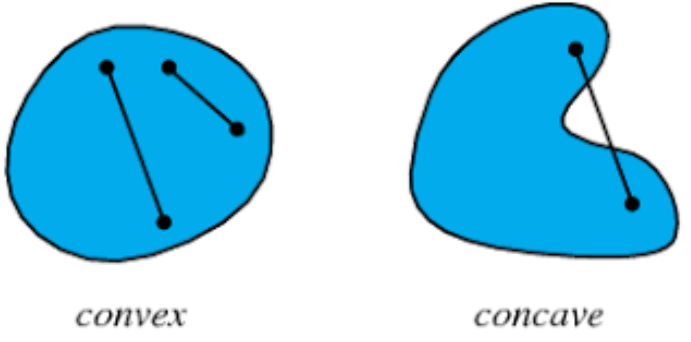
Concept/Word Space

- Consider 3-d cube of “meaning”
  - e.g. small/big, cold/hot, light/heavy
  - (0, 1, 1) => small hot heavy thing
  - (.5, 0, 0) => midsized cold light thing
- Words, phrases, even “ideas” are 3-D points
- Number of corners (words) for dimension:
  - 20?
  - 265?
- GPT uses 12,800 dimensions. Lots of “meaning” possible.

Pattern Space Regions

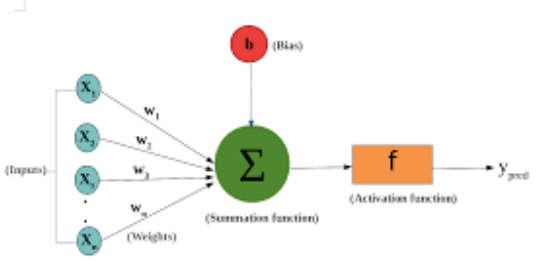
- 2-number pattern of size/weight.
  - Categories are ping-pong balls, bowling balls, balloons
- Grayscale images
  - Each pixel a dimension
  - 20x20 → 400 dimensions
  - Each digit is a region in 400-D
- Convex vs nonconvex regions
  - Is region of all 6-s convex?

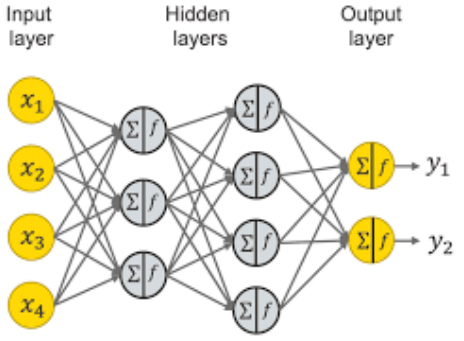




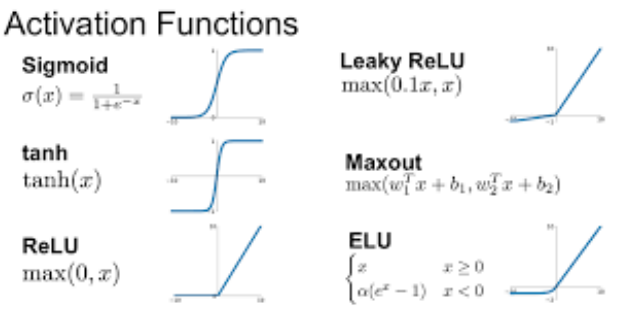
Neural Networks and Spaces

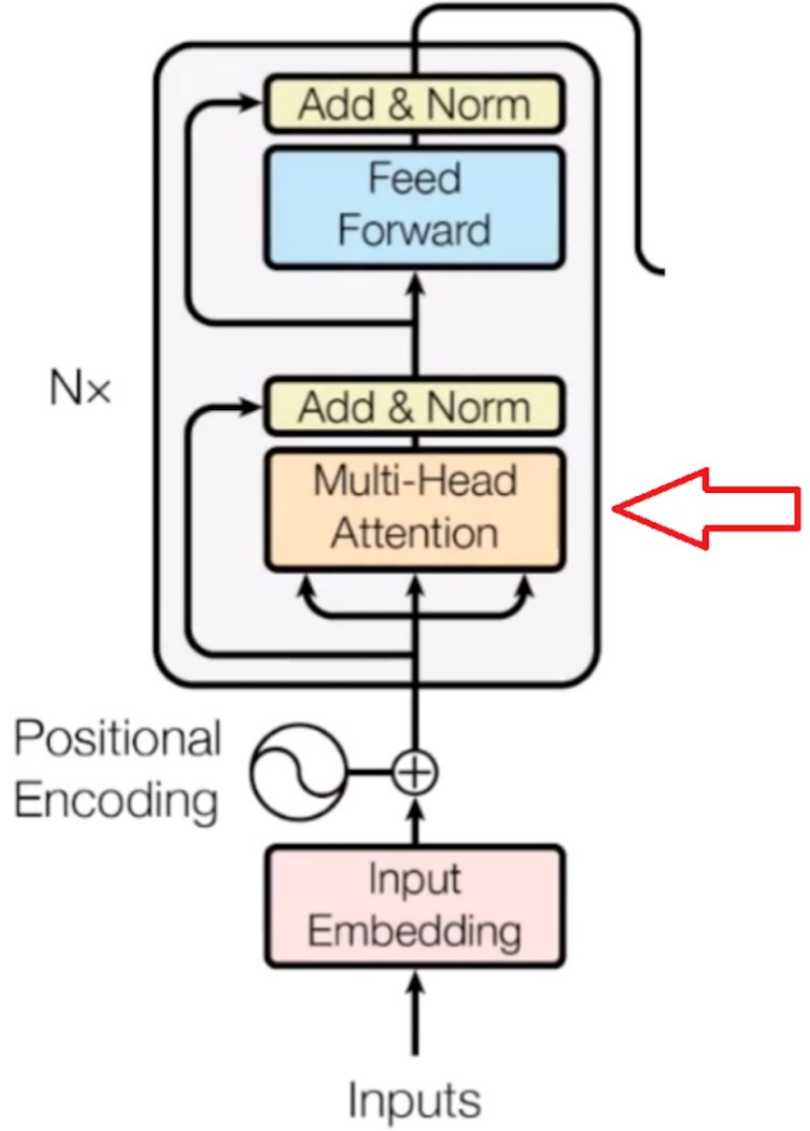
- Neuron
  - Numerical inputs from other neur
  - Weights on the inputs
  - “Activation” or triggering function





- Neural network
  - Input pattern => Layers of neurons => Output pattern
- What does it do?
  - Weighted sums are *affine transforms* in pattern space
    - Scale, rotate, translate, skew
  - Activation functions “uncrumple” the folded regions
  - Activation functions are essential to the “intelligence”



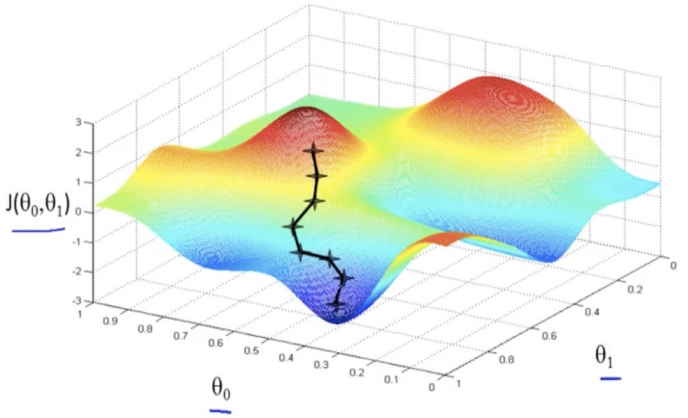

Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Nx
Positional Encoding
Input Embedding
Inputs

How GPT Learns

- A neural network has ***many*** number-weights (“parameters”)
- 175 billion parameters for GPT-3. 1.7 trillion for GPT-4.
- How do we get numbers that work? By adjusting parameters to imitate examples… Lots of examples.. 100 million sentences.
- Further refinement and “socialization” by RLHF

“Gradient Descent” Training

- Example of 2 weights
- Red mountains have high error; blue valleys have low error.
- Start with random weights, then tweak weights to move toward low error. Rinse and repeat…
- “Learning” is just rolling a ball down a hill, in high dimensions. No plan, no higher reasoning...
- The network is thus self-taught, automatically adjusted
- But, with 1+ trillion dimensions, and 100 million sentences.



The Weights *are* the “Intelligence”

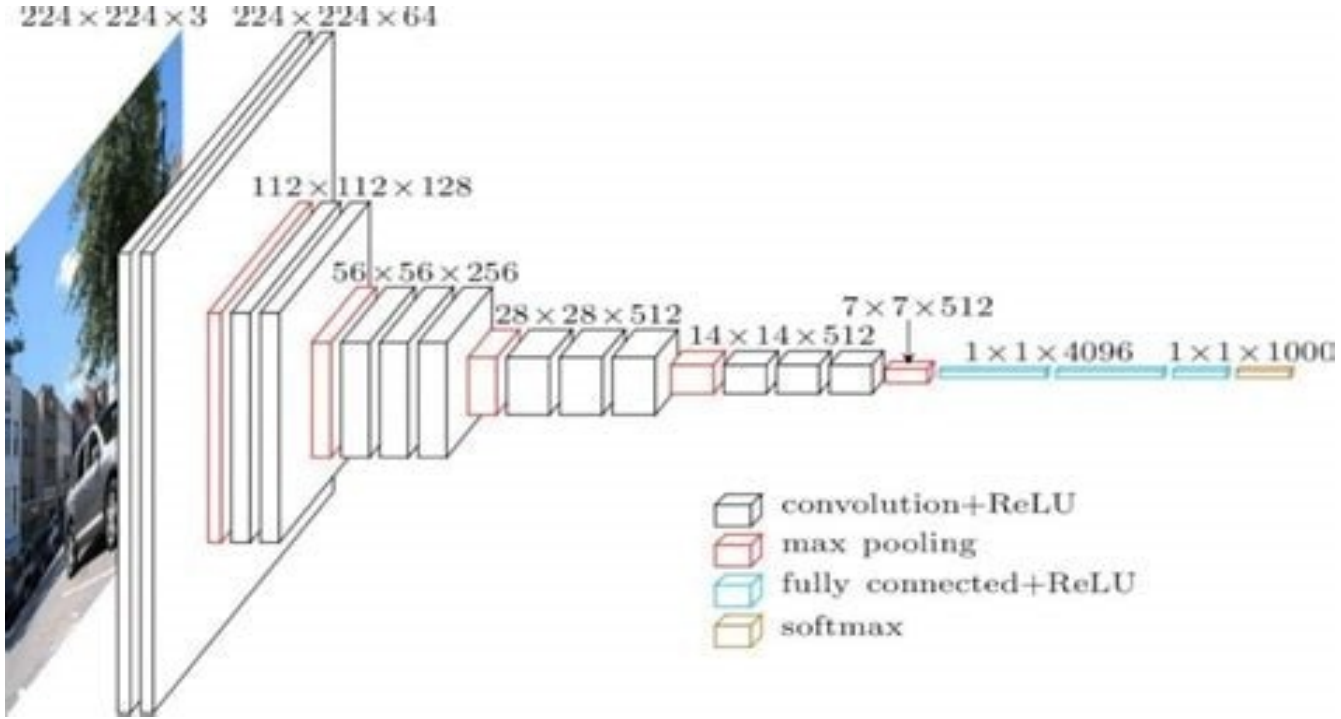
- Doesn’t need to do web hunts. Could run on a disconnected computer.
- Trained weights are “long term memory”. Current conversation is “short term memory”. All internal logic is by word/concept embeddings.
- And we *don’t know* what each weight does. It’s *self-organized* to follow examples.
- Could learn to converse in a language none of its programmers understand.
- Are world-models, theory of mind, etc. spontaneously emerging from the training?

An LLM

- Is *Not* just an improved search engine
- Is *Not* hand-crafted code
- Is *Not* the whole modern AI story
- *Is* a sophisticated mathematical engine
- *Is* self-organizing
- *Is* a partial mystery even to its creators

Other AI – CNNs

- Convolutional Neural Networks



CNN Concepts

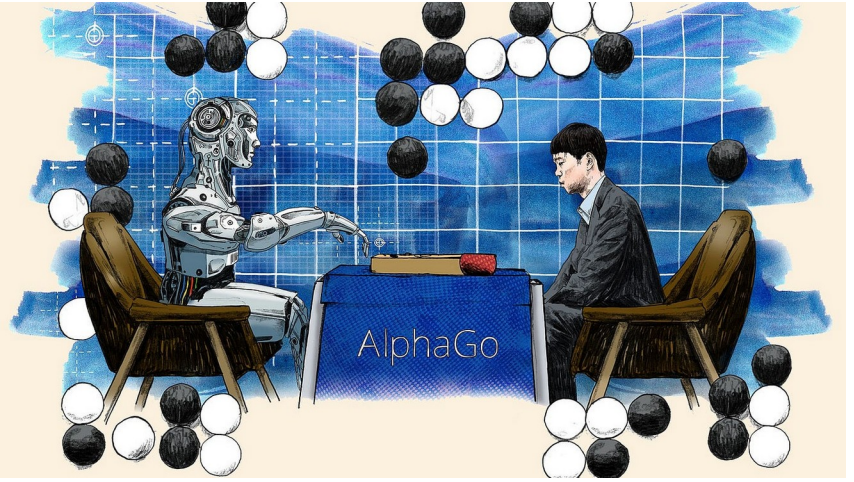
- Deep Learning, but designed for images
- Modeled after human visual system
- Trained on huge image collections
- Faces, objects, self-driving cars, etc.
- Related work in neural radiance fields (“NeRF”s) reconstructs 3-D scenes.

Other AI – Reinforcement Learning

- Automatic generation of training data
- Example of playing games against yourself to learn.
- Problem config => best move
- But, when you win, how do you know which moves helped?
- And, as you play, do you try new approaches or refine existing ones?

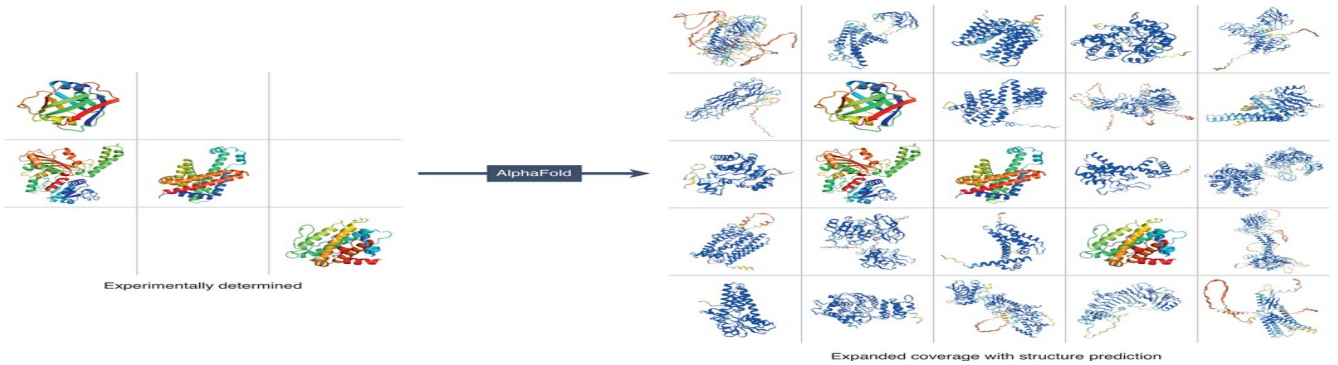
RL and Games

- Game of Go is notoriously hard. Many moves, much lookahead needed.
- Alpha-Go beats best human players using RL. And finds new strategies.



RL and Protein Folding

- GATACAGA…. => shape of protein
- Very hard, and critical to biomed research. One protein might take years of research
- Alpha-fold learns to fold via RL, and folds 100,000 proteins.



Diffusion Models

- DALL-E examples
- Form of deep learning trained on labelled images
- Translates text to image
- Can also fill in missing parts of images

AI Technical Concepts

- Word Embedding
- Pattern Space
- Gradient Descent
- Attention
- Transformer
- RLHF: Reinforcement Learning by Human Feedback
- Convolutional Neural Network (CNN)
- Reinforcement Learning

How Far Will It Advance?

- "Heavier-than-air flying machines are impossible" Lord Kelvin 1895
- It will “plateau”
  - Reaching limits on compute and on data
  - It’s plateaued before
- It will reach human intelligence or more
  - Unprecedented research community and infrastructure
  - Hardware is rapidly improving
  - Could do self-instruction via reinforcement learning

So, which is it?

- 5 years ago, most researchers would guess “plateau”. Now most would guess major improvements
- Hinton’s insight on deep learning vs. brains
  - Example of early steam power, electricity, chemistry
  - Birds vs planes
  - GPT-4 is a mouse brain
- Desperate need for mathematical theory
  - Slow but worthwhile progress on this
  - Some of the most important math of the 21st century